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Bay-head deltas across the northern Gulf of Mexico back step in response to the 8.2 ka cooling event

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ABSTRACT

Estuaries along the northern Gulf of Mexico show rapid landward shifts in estuarine environments (back-stepping) around 8200 cal yr BP. Paleogeographic maps of the estuaries based on seismic data and cores, and constrained chronologically by radiocarbon dates, depict bay-head delta retreat in excess of 25 m/yr. During these events the sub-aerial delta plain flooded to create sub-tidal upper- and middle-bay environments. Abrupt sea-level rise associated with the 8200 cal yr BP climate-change event caused these estuaries and others around the world to reorganize abruptly. In the northern Gulf of Mexico, the sea-level rise associated with the event was only ~0.1 m over a sub-decadal time period. This rise was superimposed on a background decelerating rate of sea-level rise of ~1.0–0.5 m/century. The widespread bay-head delta back-stepping, documented here, may be an appropriate analog for possible abrupt future climate change in response to global warming. It also shows that estuaries fringed by low-gradient shorelines are highly sensitive to low-amplitude and sudden sea-level rise.

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1. Introduction

Rapid landward translations of estuarine environments during the early Holocene are commonly linked to rapid sea-level rise events. These back-stepping events are preserved in the stratigraphic record as a retrogradational parasequence set (Van Wagoner et al., 1990; Rodriguez et al., 2005). In the northern Gulf of Mexico, Wilkinson and Byrne (1977) and Thomas and Anderson (1994) identified early Holocene estuarine back-stepping events in the Lavaca and Trinity incised-valley-fill successions, respectively. Although an increase in the rate of sea-level rise was inferred as the forcing mechanism, the underlying cause for the sea-level jump was unknown because of poor correlation of the back-stepping events with existing sea-level curves. It is possible that estuarine environments could back step in response to an increase in the rate of sea-level rise that is below the resolution of current early Holocene sea-level curves (±1.0 m; Törnqvist et al., 2004; Milliken et al., 2008a). Additionally, these earlier studies lacked control on other back-stepping forcing mechanisms such as autoretreat (Muto and Steel, 1992) or antecedent topography (Rodriguez et al., 2005).

Vaughan et al. (2007) have emphasized that decadal to century-scale sea-level rises of decimeter magnitude are expected from episodic and localized retreat of ice sheets, such as the collapse of individual ice streams. Indeed, the record that has emerged from studies of different ice streams in West Antarctica is that their retreat was diachronous (Anderson et al., 2002; Mosola and Anderson, 2006; Heroy and Anderson, 2007). Hence, the overall sea-level rise of the Holocene, especially the early Holocene when the Laurentide Ice Sheet existed, should be punctuated by short-lived events of a few decimeters magnitude. Such events would be below the resolution of most sea-level curves.

Hori and Saito (2007) recognized an estuarine back-stepping event at approximately 8500 cal yr BP in three incised-valley-fill systems in Asia: the Changjian, the Song Hong, and the Kiso River. They identified the back-stepping events at each site as a decrease in sediment-accumulation rates and suggest that environmental changes were caused by an increase in sea level associated with the catastrophic drainage of glacial lakes Agassiz and Ojibway, marking the final collapse of the Laurentide Ice Sheet around 8400 cal yr BP (Alley et al., 1997; Barber et al., 1999; Teller et al., 2002; Clarke et al., 2003; Ellison et al., 2006; Wiersma and Renssen, 2006; Hori and Saito, 2007). Kendall et al. (2008) created a numerical model that shows sea-level rise due to the lake-drainage event around Asia, was 0.4–1.0 m. This translates into a rapid rate of rise considering the event likely lasted less than 10 yr (Barber et al., 1999; Alley and Ágústsdóttir, 2005). Given the high-amplitude and short-duration of the sea-level rise event on this part of the Earth, the dramatic

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impact to widespread coastal systems described by Hori and Saito (2007) should be expected.

Determining whether or not low-amplitude (decimeter scale) and short-duration (centennial to decadal) sea-level rise events have a perceptible impact on estuaries will help evaluate their sensitivity to similar future sea-level rise. Along the northern Gulf of Mexico, the rapid drainage of glacial lakes Agassiz and Ojibway produced only ~0.1 m sea-level rise over the sub-decadal time period (Kendall et al., 2008). The amplitude discrepancy between this area and Asia is the closer proximity of the northern Gulf of Mexico to the glacial lakes, where the gravitational attraction and resulting ocean tilt disappeared as the lakes drained (Törnqvist et al., 2004; Alley and Ágústsdóttir, 2005; Kendall et al., 2008). Bay-head deltas are sensitive to small changes in accommodation, which are strongly controlled by sea level, making this depositional environment ideal for examining the impact from the 8200 cal yr BP event. This sensitivity is because bay-head deltas have extremely low gradients along depositional dip and are confined within flooded paleo-valleys. These characteristics should force a bay-head delta to shift landward rapidly (back step) in response to a small increase in accommodation, but is a sudden 10 cm rise in sea level large enough to cause bay-head delta back-stepping?

The focus of this study is on the upper-estuarine response to sea-level rise between 8800 and 7800 cal yr BP along the northern Gulf of Mexico. During this time period, long-term sea level at the site, of which the sea-level pulse from drainage of the glacial lakes at 8400 cal yr BP is superimposed upon, was decelerating from ~1.0 to 0.5 m/century (Törnqvist et al., 2004; Kendall et al., 2008; Milliken et al., 2008a). The spatial distribution of paleo-depositional environments is easy to interpret from the stratigraphic record of wave-dominated estuaries because distinct physical processes and lithologic facies characterize the three main depositional environments, including the barrier island complex, which incorporates tidal inlets and deltas, the middle bay (a.k.a. central basin), and the bay-head delta (Dalrymple et al., 1992). Estuaries with the highest potential of preserving a nearly complete sedimentary record are those characterized by deep-basal incisions (high accommodation) and low rates of post-depositional reworking by waves, tides, and organisms. Mobile, Sabine, Galveston, Corpus Christi, and Baffin bays were chosen for the study because each bay is located in a different climate setting and the bays have drainage areas of various sizes, are in microtidal settings (tidal amplitude less than 1.0 m) and have well-constrained stratigraphies (Figs. 1 and 2). Looking for contemporaneous flooding surfaces in more than one estuary whose climate setting, discharge, and valley shape vary along the coast, allows us to isolate sea-level rise as the back-stepping forcing mechanism.

2. Methods

To map incised-valley morphology and paleo-depositional environments in each estuary, high-resolution single-channel seismic data (1.5 kHz boomer and 2–15 kHz chirp sources) and cores were collected, compiled from previous studies, and examined. Details on data-collection methods for Mobile, Sabine, Galveston, and Corpus Christi Bays are presented in Anderson and Rodriguez (2008) and for Baffin Bay in Simms et al. (2010). Maps of the antecedent topography, the surface inundated with rising sea level (exposure surface), are important to constrain for each estuary because morphological variations strongly control sediment accommodation and back-stepping events (Belknap and Kraft, 1985; Fletcher et al., 1990; Bratton et al., 2003; Rodriguez et al., 2004, 2008). Maps of this surface in Mobile (Kindinger et al., 1994; Greene et al., 2007), Sabine (Anderson et al., 1991; Milliken et al., 2008b), Galveston (Smyth et al., 1988; Thomas and Anderson, 1994), Corpus Christi (Wright, 1980; Durbin et al., 1997; Kindinger et al., 2003; Simms et al., 2008) and Baffin (Simms et al., 2010) bays were incorporated into this study. Using a small barge with a hydraulic rotary-drill rig, cores ranging from 20 to 14 m in length were collected from the estuaries. Cores are not continuous and flooding surfaces, defined as the contact separating younger from older strata across which there is evidence of an abrupt increase in water depth, were targeted for sampling. Flooding surfaces were initially recognized and mapped from seismic data interpretations and these maps were used to select coring sites. The axis of the paleo-valley was selected as the coring target in each estuary because that location ensures the longest sedimentary record.

Multiple flooding surfaces were recognized in each estuary and the surface closest to ~14–13 m below mean sea level (bmsl), which is the level of the sea in the study area ~8200 cal yr BP (Toscano and Macintyre, 2003; Törnqvist et al., 2004; Milliken et al., 2008a) was targeted. Beta Analytic, the University of Arizona, and the Woods Hole Oceanographic Institution radiocarbon dated (accelerator mass spectrometry) shell and organic material sampled from the cores (Table 1). Ages (29 dates) were calibrated to calendar years using the Calib 5.0 northern hemisphere terrestrial

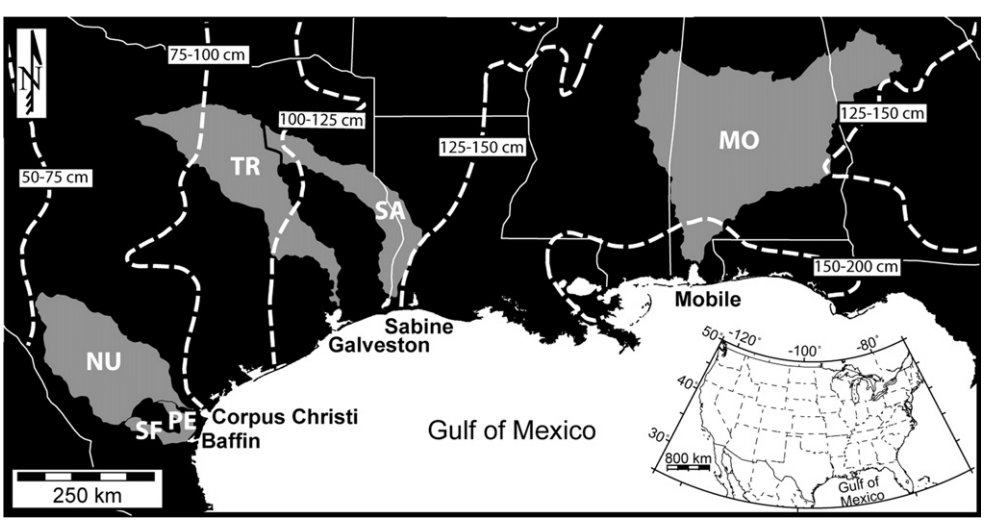

Fig. 1. Study area including the drainage basins of the rivers forming the five estuaries. Isopleths of mean annual precipitation (dashed lines) increase to the east. River abbreviations are: MO – Mobile, SA – Sabine, TR – Trinity, NU – Nueces, LO – Los Olmos, SF – San Fernando.

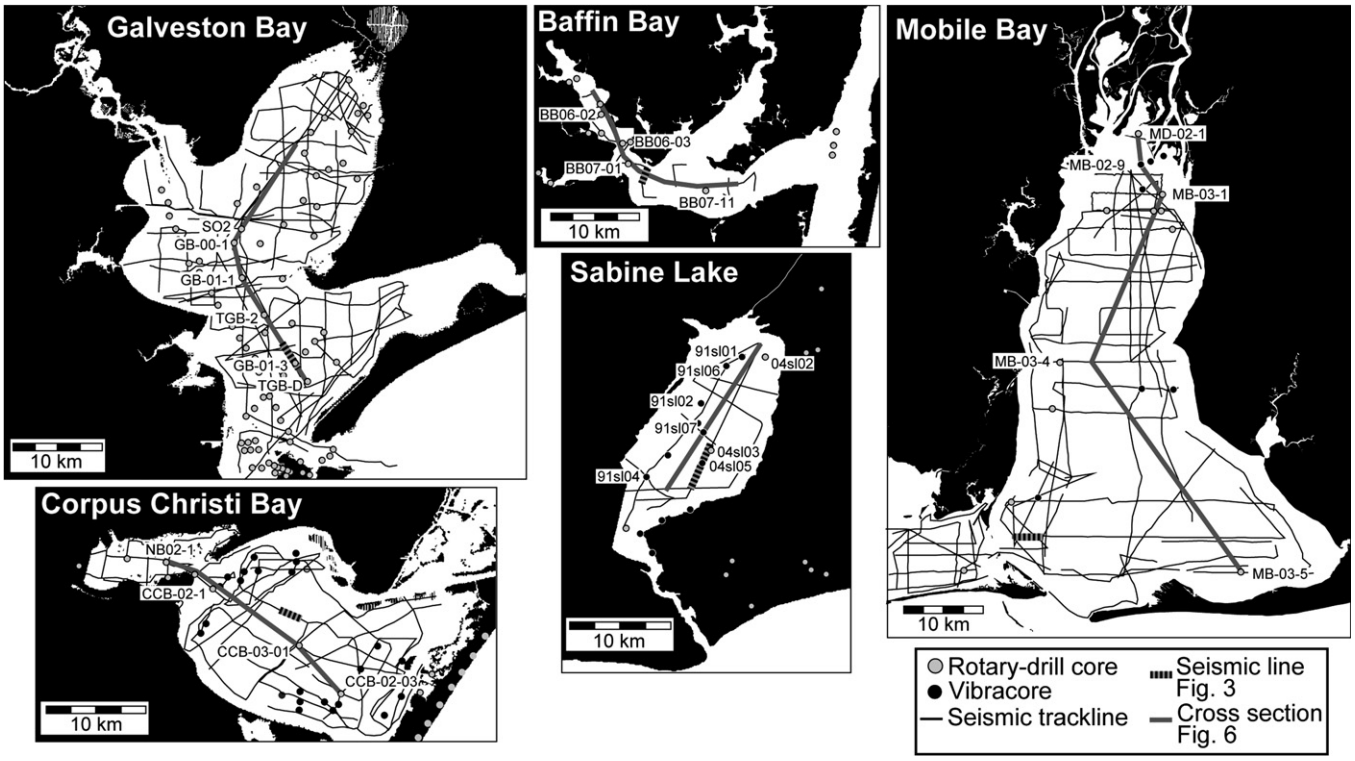

Fig. 2. Maps of the estuaries showing the dataset used in this study and locations of subsequent figures.

curve (Stuiver and Reimer, 1993) after applying a correction for the local radiocarbon reservoir. The local radiocarbon reservoir was only applied to shell material (not wood or peat) and was determined by paired shell/wood ^{14}C age determinations from the published literature (e.g. Aten, 1983; Table 1). It is essential that material being dated is preserved in situ. Whenever possible, individual wood or plant fragments and articulated bivalves were chosen for dating over bulk peat samples and unpaired valves. The timing of the flooding surface is constrained by dating material sampled above and below the contact, as close to the contact as possible. Ideally, the flooding surface would have been dated at multiple locations up the axis of the estuary, but suitable material for dating did not always exist at the sampled sites. Multiple ages from various depths were obtained to ensure that dated materials were not reworked. Sample depths are not corrected for autocompaction because cores were not continuous and some dewatering occurred during core cutting.

3. Depositional environments and estuarine stratigraphy

Anderson et al. (2004) and Simms et al. (2007) mapped a regional exposure surface that correlates with the Oxygen Isotope Stage 2 sea-level lowstand on the northwestern Gulf of Mexico shelf. The depth of this surface correlates with maps of the exposure surface in Mobile, Sabine, Galveston, Corpus Christi, and Baffin bays. Generally, the exposure surface is imaged with seismic data as a high-amplitude reflector that shows erosional truncation of reflectors below and onlap above (Fig. 3). The surface has high-relief (>25 m) and this antecedent topography was flooded during the last deglaciation forming the estuaries. The exposure surface is always sharp in cores and is commonly characterized by an increase in sediment shear strength, the presence of calcareous nodules, and a change in color down core from gray, to moderate reddish brown, moderate yellowish brown, light brown green, or yellow (Figs. 4 and 5a). The sediments below the exposure surface are interpreted as alluvial or open-marine and estuarine deposits, which were likely deposited after and before Oxygen Isotope Stage 3, respectively (Morton et al., 1996).

The estuaries contain an overall transgressive sequence above the exposure surface. Initial inundation of modern estuaries occurred no earlier than 9600 cal yr BP based on dates obtained directly above the exposure surface in Mobile Bay (Greene et al., 2007), Sabine Lake (Milliken et al., 2008b), Galveston Bay (Rehkemper, 1969; Rodriguez et al., 2005), Corpus Christi Bay (Simms et al., 2008), and Baffin Bay (Simms et al., 2010). Each wave-dominated estuary is typical of the classic transgressive sequence highlighted in Dalrymple et al. (1992), which includes bay-head delta facies overlain by upper- or middle-bay facies, and in the seaward portions of the bay, those two facies are overlain by barrier island, tidal-delta, and washover deposits (Anderson and Rodriguez, 2008; Simms et al., 2010). Overlaying the exposure surface, a seismic facies with acoustically laminated to inclined (shingled) reflections exists (Fig. 3). In Mobile Bay, Sabine Lake, and Galveston Bay, this seismic facies is a carbonaceous mud unit with abundant *Rangia cuneata*. These bivalve and lithofacies are common to the distributaries and mouth bars of modern bay-head deltas of the northern Gulf of Mexico (Parker, 1960; Figs. 4 and 5). Within Corpus Christi and Baffin bays, the seismic facies is represented by an organic-rich mud with prevalent plant fragments and possible root traces, interpreted as a delta plain environment, or a poorly sorted medium-grained sand unit, interpreted as a channel-mouth bar environment (Figs. 4 and 5c). The upper surface of the bay-head delta is easily identified through most of the estuaries with seismic data as enhanced reflection, which, in some places, completely masks the underlying seismic record (e.g., Fig. 3c). This is attributed to gas, which is likely methane produced by the bacterial degradation of organic matter in the carbonaceous clay unit. Bubbles are commonly observed breaking the surface of the water as the top of the bay-head delta is penetrated during coring.

In Sabine Lake, Corpus Christi Bay, and Baffin Bay the bay-head delta grades seaward into a seismic facies characterized by high-

Table 1
Radiocarbon dated samples from the five estuaries and information on corrections applied and data sources.

Lab code	Sample ID	Material	^{14}C age	Error	Reservoir applied	Median calendar age[d]	2-σ Range	Environment	Source
Mobile Bay, Alabama									
OS-40279	MD-02-1 1229	Wood	7190	35	0	8000	8060–7940	Delta plain	Rodriguez et al. (2008)
OS-37367	MD-02-1 1310	Wood	8000	55	0	8850	9010–8690	Delta plain	Rodriguez et al. (2008)
OS-41176	MB-03-05 1368	*Corbula contracta*[a]	7680	40	Marine	8140	8020–8260	Middle bay	Rodriguez et al. (2008)
OS-44826	MB-03-05 1412	Wood	8060	40	0	8990	9090–8780	Delta plain or Marsh	Rodriguez et al. (2008)
OS-42942	MB-03-04 1514	Peat	8270	60	0	9260	9440–9030	Delta plain or Marsh	Rodriguez et al. (2008)
OS-42768	MB-03-01 1095	Wood	6980	35	0	7820	7930–7710	Middle bay	Rodriguez et al. (2008)
Sabine Lake, Texas/Lousiana									
OS-46747	04sl02_30_35_18_21	Peat	7110	45	0	7940	8010–7850	Delta plain	Milliken et al. (2008)
OS-46764	04sl02_30t35_76	Peat	7240	30	0	8050	8160–7980	Delta plain	Milliken et al. (2008)
OS-46762	04sl03_45_50_2	Peat	7180	30	0	7990	8040–7950	Upper bay	Milliken et al. (2008)
22976	04sl03_45t50_129	Wood	7670	30	0	8450	8540–8410	Delta plain	Milliken et al. (2008)
Galveston Bay, Texas									
AA44772	GB-01-3 1507-1511	Peat	7666	70	0	8470	8590–8370	Delta plain	Anderson et al. (2008)
AA44771	GB-01-3 1252-1254	*Mulinia lateralis*	7275	100	225	7870	8040–7670	Middle bay	Anderson et al. (2008)
OS-73347	GB-01-1 1406-1410	*Rangia cuneata*	7820	35	225	8400	8450–8350	Delta plain	Unpublished
AA44774	GB-01-1 1460-1464	Wood	7803	70	0	8590	8950–8420	Delta plain	Anderson et al. (2008)
OS-34715	GB-00-1.1 1026-1029	Wood	6900	35	0	7730	7830–7670	Delta plain	Anderson et al. (2008)
AA39384	GB-00-1.2 1126-1133	Wood	7413	53	0	8250	8370–8060	Delta plain	Anderson et al. (2008)
39032	GB-00-1.2 1288-1298	Wood	7660	40	0	8450	8540–8390	Delta plain	Anderson et al. (2008)
N/A	TGB-D 16.46	*Rangia cuneata*	7968[b]	290	225	9060	9730–8370	Delta plain	Rehkemper (1969)
Corpus Christi Bay, Texas									
OS-42911	CCB03-01 15.44	*Nuculana concentrica*[a]	7890	60	760	7960	8150–7800	Middle bay	Simms et al. (2008)
OS-42912	CCB03-01 16.29	*Abra aequalis*[a]	8050	40	760	8100	8180–8020	Upper bay	Simms et al. (2008)
36754	CCB02-01 12.6	*Mulinia lateralis*	7750	50	760	7830	7930–7710	Middle bay	Simms et al. (2008)
36755	CCB02-01 15.62	*Mulinia lateralis*	8750	40	760	8870	9010–8660	Delta Plain	Simms et al. (2008)
OS-41659	NB02-01 10.28	*Nuculana concentrica*[a]	7060	60	760	7730	7420–7020	Delta Plain	Simms et al. (2008)
OS-41660	NB02-01 12.07	*Barnicule*[a]	8000	65	760	8070	8180–8000	Delta Plain	Simms et al. (2008)
Baffin Bay, Texas									
OS-73348	BB-07-01 1394–1395	*Anomalocardia cunemeiris*	8440	50	200	9210	9400–9030	Middle bay	Unpublished
MTC-08902	BB07-01 1113 cm	*Mulinia lateralis*[a]	7370[c]	130	200	8000	8300–7700	Delta Plain	Simms et al. (2010)
MTC-12310	BB07-11 1543 cm	*Mulinia lateralis*[a]	6950[c]	115	200	7610	7830–7430	Middle bay	Simms et al. (2010)
MTC-12311	BB07-11 1690 cm	*Mulinia lateralis*[a]	8030[c]	115	200	8670	8990–8420	Middle bay	Simms et al. (2010)
MTC-12312	BB07-11 1727 cm	Unknown shell	8540[c]	120	200	9320	9530–9030	Middle bay	Simms et al. (2010)

[a] Articulated.
[b] Radiometric date from Rehkemper (1969); Assumed a d^{13}C value of 0.0.
[c] See Simms et al. (2010) for details on ^{13}C corrections.
[d] Using the Northern Hemisphere Curve of Reimer et al. (2004).

amplitude acoustically laminated reflectors. This seismic facies is sampled as a fine- to medium-grained sandy mud/muddy sand unit with discrete and irregular layers of carbonaceous material and is interpreted as upper bay. Fig. 3b clearly shows the stratigraphic relationship between bay-head delta sub-environments and upper-bay facies in Sabine Lake, where the prodelta is imaged as seaward-dipping sigmoidal reflectors interpreted as clinoforms that downlap and grade seaward into the upper-bay seismic facies. Above the bay-head delta or upper bay is a seismic facies represented by acoustically transparent to laminated seismic reflectors. This seismic facies is sampled as a homogeneous gray mud unit with abundant *Crassostrea virginica* and is interpreted as middle bay. The middle-bay facies of all five bays is very similar. In Corpus Christi Bay, mounded high-amplitude reflectors, which are onlapped by the continuous reflectors of the middle-bay seismic facies, exist directly on the flooding surface between upper and middle bay (Fig. 3d). These mounds are interpreted as oyster reefs that colonized as salinity increased across the flooding surface. Cores collected adjacent to these mounds sampled fragments of *C. virginica* and *Ostrea equestris* at depths similar to the mounds (Simms et al., 2008). In the distal area of all of the estuaries, middle-bay inter-fingers with a seismic unit characterized by chaotic to bidirectional dipping reflectors sampled as sand to shelly sand and interpreted as flood-tidal-delta and washover deposits (e.g., Fig. 3c).

The vertical succession of facies within the present geographical center axis of each estuary shows a very similar architecture (Fig. 6). The underlying bay-head delta deposits (or the exposure surface in many locations in Mobile Bay) are in sharp contact with the overlying estuarine deposits in both cores (Figs. 4 and 5) and seismic profiles (Fig. 3). Missing in the vertical succession through large portions of Mobile and Galveston bays is the transitional upper-bay facies found in modern dip profiles through all of the bays. This depositional environment shows higher preservation in Sabine Lake, Corpus Christi Bay, and Baffin Bay. Compared to today, Mobile Bay and Galveston Bay were likely larger around 8200 cal yr BP because their associated barrier islands were located seaward on the low-relief continental shelf (Rodriguez et al., 2004; Rodriguez and Meyer, 2006). This is well constrained for Galveston Bay where Rodriguez et al. (2004) show that the Bay was greater than twice its present size ~8200 cal yr BP and extended onto the inner continental shelf. Given greater fetch length, bay-ravinement processes (i.e. tidal-, wave- and current-erosion as sea-level rises) in Galveston and Mobile bays were likely more erosive around 8200 cal yr BP than today. Corpus Christi Bay was likely smaller 8200 cal yr BP than today because Mustang Island, an aggradational barrier island that isolates Corpus Christi Bay from the Gulf of Mexico, was at its present location at that time (Simms et al., 2006). Higher preservation of the upper-bay depositional environment in

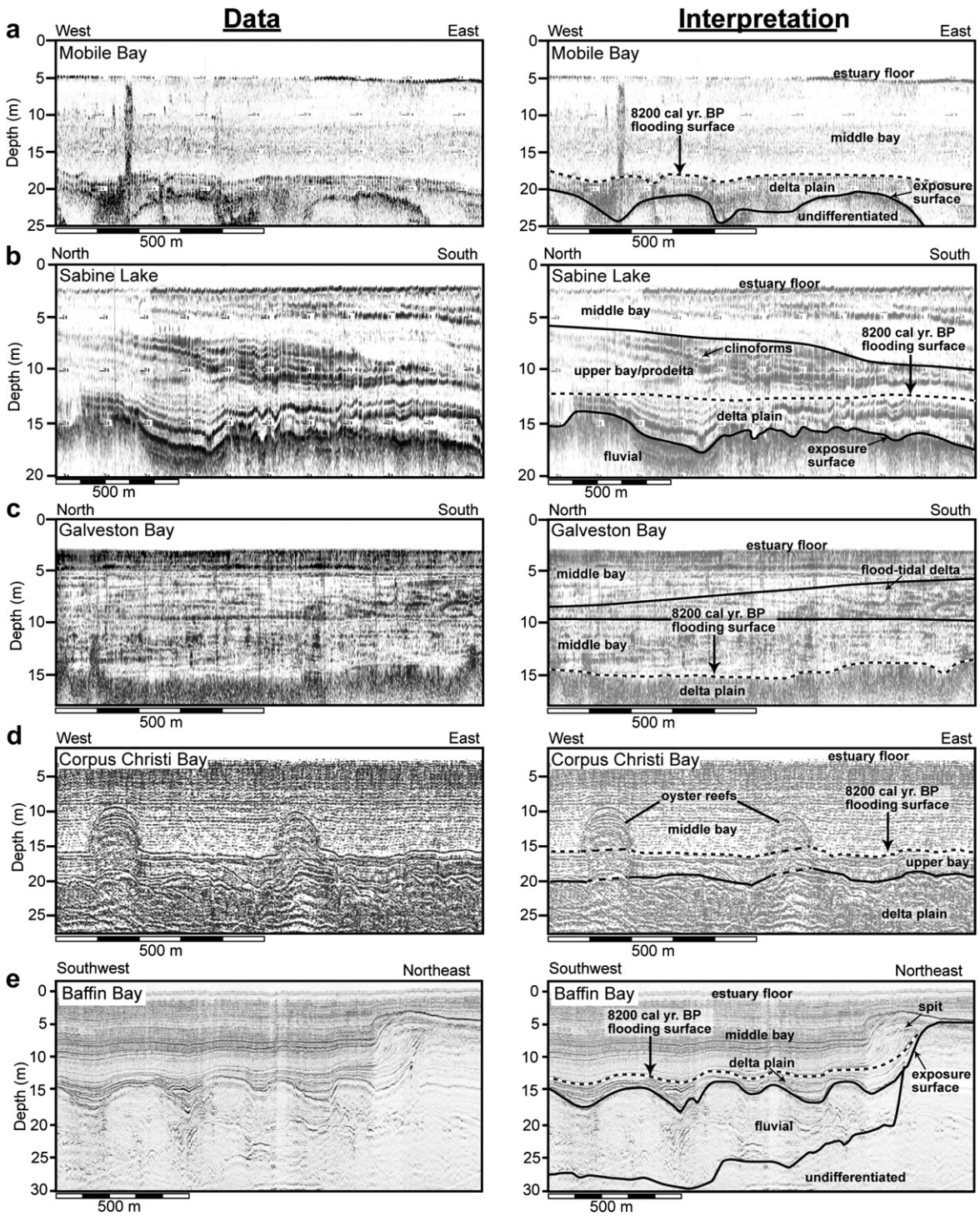

Fig. 3. The seismic profiles from each of the estuaries show the 8200 cal yr BP flooding surface separating proximal from distal depositional environments, each characterized by a distinct seismic facies. Time-to-depth conversion was made using 1500 m/s and 0 m approximates mean sea level. See Fig. 2 for locations.

Baffin Bay, Corpus Christi Bay, and Sabine Lake is likely because these estuaries were more restricted and smaller than Mobile Bay and Galveston Bay, making bay-ravinement processes less erosive.

4. Timing of flooding surface formation and estuarine paleogeography

The depositional sequence in the estuaries is characterized by proximal units, like bay-head delta and marsh environments, separated from the more distal upper- and middle-bay environments by a discrete flooding surface. Seismic data clearly images the contacts between upper and middle bay and underlying more proximal environments as a high-amplitude reflector and/or a change in acoustic facies (Fig. 3). The flooding surface in Sabine Lake (~13 m bmsl), Galveston Bay (~14 m bmsl), Corpus Christi Bay (~18 m bmsl) and Baffin Bay (~14 m bmsl) is mapped as being flat and extensive. The flooding surface in Mobile Bay (~14 m bmsl) is the most extensive, extending for over 30 km. The flat

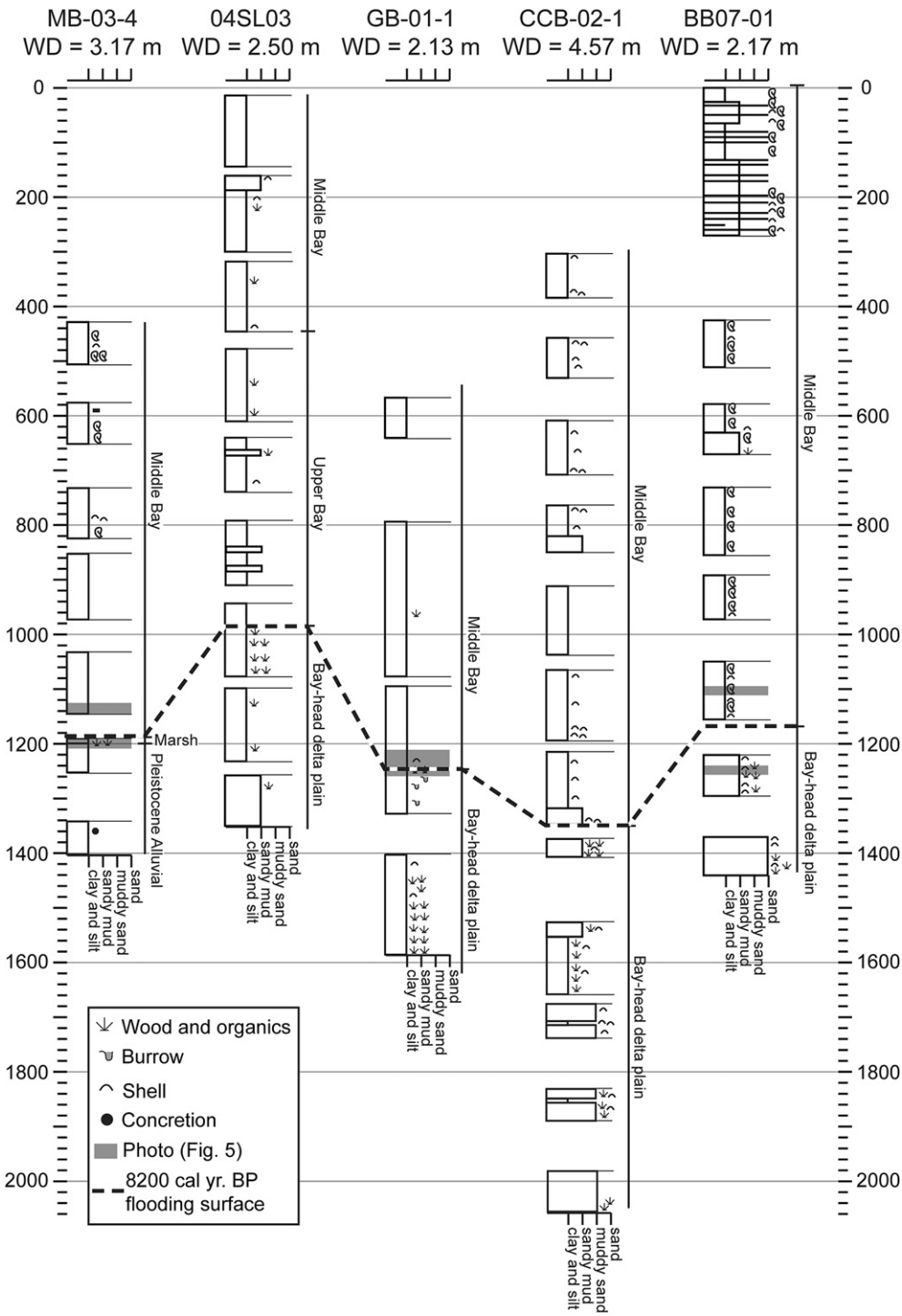

Fig. 4. Core logs showing lithologic facies above and below the flooding surface. Missing intervals represent depths that were not sampled. Zero meter corresponds to the bay floor and WD = water depth. See Fig. 2 for locations and Table 1 for information on dates.

morphology and large extent of the flooding surfaces indicate that the environments changed rapidly and drastically. The similar depth of the flooding surfaces between the estuaries suggests that they formed at the same time and radiocarbon dates below and above the surface in the estuaries support this inference showing dates between 9210 and 8100 cal yr BP and between 8140 and 7820 cal yr BP, respectively (Fig. 6, Table 1). Only one date below the flooding surface yielded an age younger than 8200 cal yr BP (8180–8020) and this was from upper-bay sediment in Corpus Christi Bay (Fig. 6; Table 1). A sample from delta plain sediment at the same depth, about 10 km further up dip, yielded an age of 8870 cal yr BP, which suggests that the younger, more distal bivalve (an articulated *Abra aequalis*) may have burrowed below the flooding surface before death; it was sampled <10 cm below the flooding surface. Although the flooding surfaces are sharp in terms of lithologic and acoustic change below and above the contacts, appropriate material for dating was typically found within 50 cm above and below the surfaces making the measured hiatus across it,

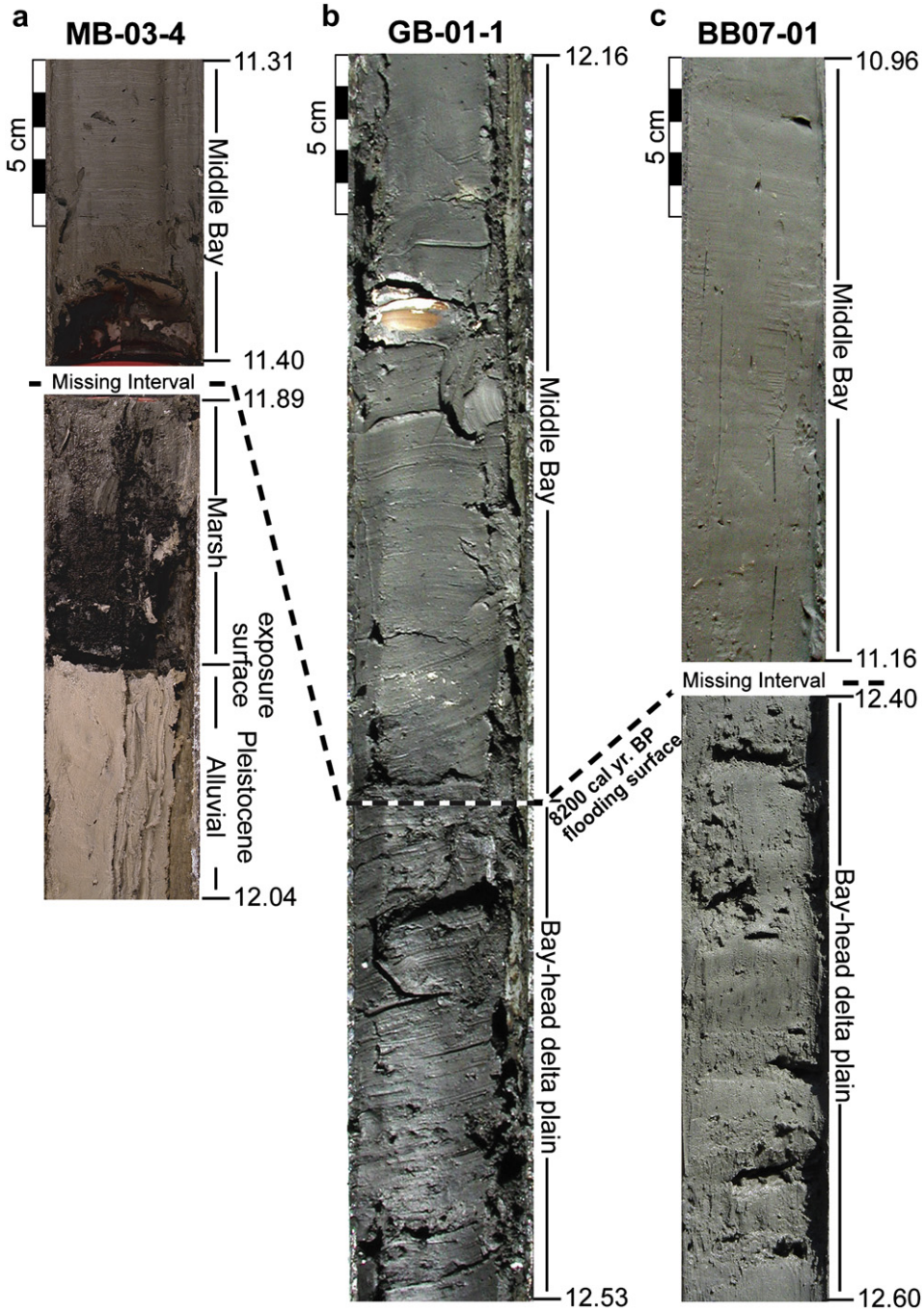

Fig. 5. Photographs of cores from Mobile Bay (a), Galveston Bay (b) and Baffin Bay (c) show the sharp nature of the 8200 cal yr BP flooding surface and the sediments above and below. Depths are in m relative to the bay floor. See Fig. 2 for core locations and Fig. 4 for photographed intervals.

ranging from about 140 to 1210 yr, larger than the true hiatus associated with the landward movement of the depositional environments. In addition to measurement error, which includes analytical error and the much larger error associated with sampling appropriate material for quantifying the hiatus, the duration of the hiatus is also controlled by bay-ravinement processes removing material from the underlying and more proximal depositional environments as water depth increases.

Since the flooding surface imaged with seismic data is easy to recognize and always correlates to a sharp contact between depositional environments in the cores, the seismic data sets were used to map the paleogeography of the estuaries below and above the flooding surfaces (Fig. 7). In all of the estuaries studied, a dramatic change in the distribution of depositional environments (e.g. bay-head delta and middle-bay environments) occurred across the flooding surfaces and the measured hiatus' includes the 8200 cal yr BP climate-change event. Bay-head deltas moved landward >40 km in Mobile Bay, >15 km in Sabine Lake, Galveston Bay, and Corpus Christi Bay, and >10 km in Baffin Bay. Bay-head deltas show an aggradational geometry at their new landward position, which indicates a period of stability in the position of the delta front depositional environment. Based on the hiatus across the flooding surface, these changes took place in less than 800 yr for Mobile Bay, 400 yr for Sabine Lake, 500 yr for Galveston and Corpus Christi

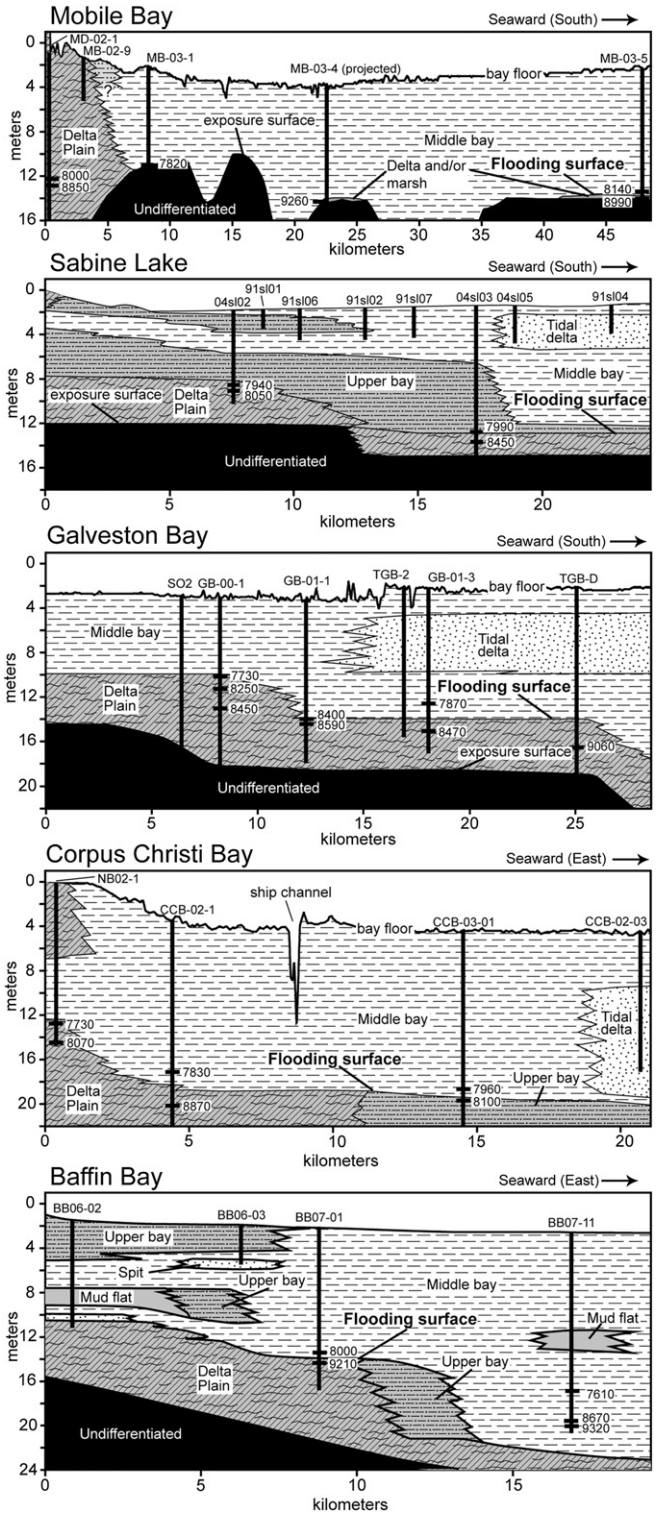

Fig. 6. Cross-sections from seismic and lithologic data oriented close to the axis of the paleo-valleys. High-relief topography of the exposure surface in Mobile Bay is due to seaward-branching paleo-valleys (Greene et al., 2007). The horizontal bars through the cores are labeled as the median radiocarbon calendar age (see Table 1 for details). Zero meter approximates mean sea level. For location of cross-sections, see Fig. 2.

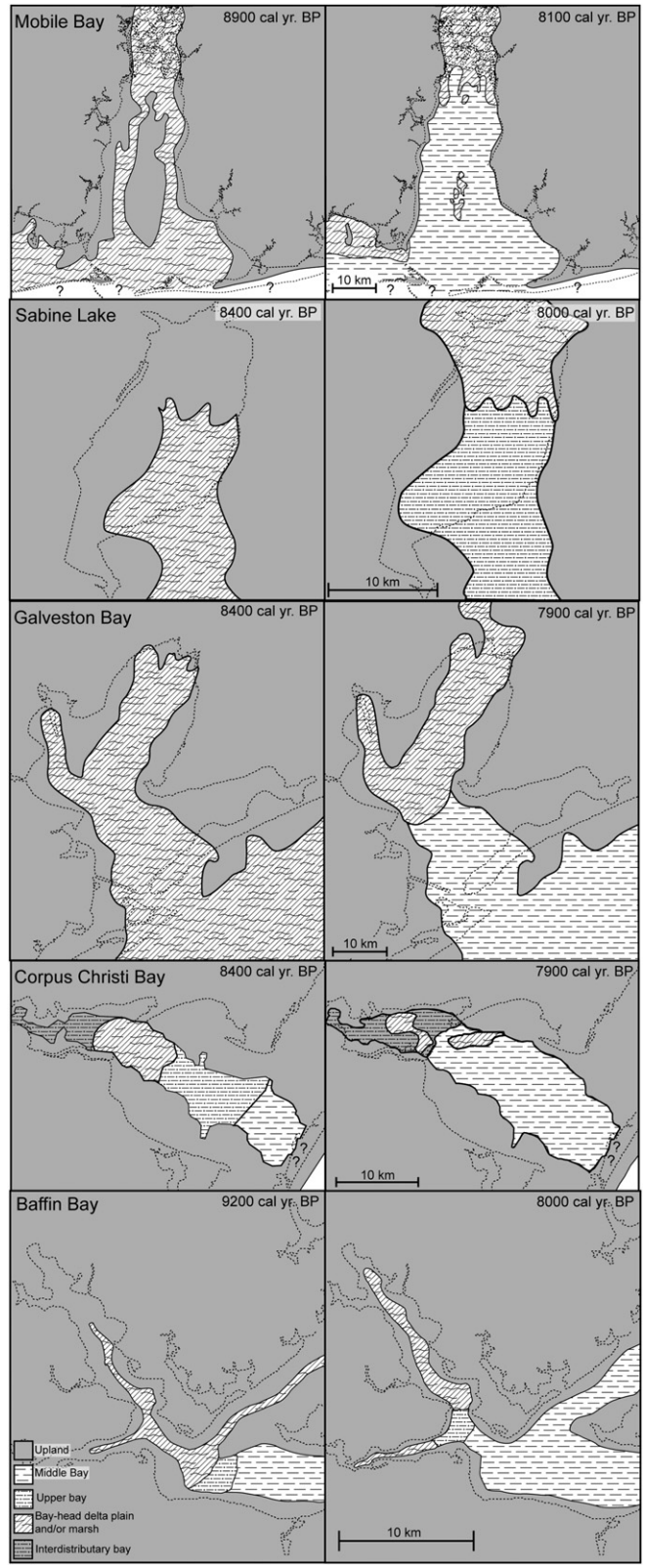

Fig. 7. Paleogeographic maps before and after back-stepping based on seismic data and cores. Timing is constrained by interpolating between radiocarbon dates above and below the flooding surface, making the time interval during which the environmental change took place a maximum.

Bays, and 1000 yr for Baffin Bay. Environmental changes likely occurred more rapidly than the hiatus. This is because bay ravinement removes material below the flooding surface making radiocarbon dates from these sediments older than the environmental change event (i.e. the 8200 cal yr BP climate-change event).

Although the dates above the flooding surface are all younger than the event, these sediments are not influenced by the erosion associated with the flooding surface and, therefore, should accurately constrain the timing of initial deposition associated with the new more distal environment. The difference between radiocarbon dates above the flooding surfaces in distal and proximal locations along the axis of the estuary is an indicator, albeit using an end-point method, of the rate of environmental change. Sediment directly above the flooding surface was dated in two locations along the axis of Mobile Bay, Sabine Lake, Galveston Bay, and Corpus Christi Bay and the samples are separated by 40 km for Mobile Bay and 10 km for the other estuaries. The 2-σ analytical ranges of the radiocarbon dates for all of the estuaries overlap except for Mobile bay where the ranges are separated by 90 yr, which implies that the rate of environmental change for most of the bays is within the resolution of the data (average range is 250 yr). Even if the radiocarbon dates taken above the flooding surface at landward and seaward locations along the axis of the estuaries were separated by 400 yr, that translates into bay-head delta back-stepping rates of 25 m/yr for Sabine Lake, Galveston Bay, Corpus Christi Bay, and Baffin Bay, and 100 m/yr for Mobile Bay.

5. Back-stepping forcing mechanisms

While a bay-head delta back-stepping event and flooding surface formation in one estuary could simply be the result of decreased river sediment discharge or autocyclic processes, such as bay-head delta lobe shifting or flooding of irregular antecedent topography (Rodriguez et al., 2005), linked environmental change along the entire northern Gulf of Mexico implies an allogenic factor was involved. Along many coastlines, the increase in the rate of sea-level rise would have manifested itself as a flooding surface and a landward shift in depositional environments within estuaries. If estuarine sediment supply was constant during this time period, an increase in the rate of sea-level rise would expand the area of the estuary and the space available for deposition. This would cause sedimentation rates at any single estuarine location to decrease. A decrease in the rate of sedimentation is documented in the paleo-Changjiang estuary, currently the Changjiang delta located in the East China Sea near Shanghai, from 6.3 mm/yr to 2.8 mm/yr (a 55.6% decrease) between 8800 and 8000 cal yr BP (Hori et al., 2001, 2002; Hori and Saito, 2007). The Guadalete wave-dominated estuary, located in the Gulf of Cadiz, experienced a decrease in the rate of sedimentation from 15.7 mm/yr to 3.9 mm/yr (a 75.2% decrease) between 8800 and 8000 cal yr BP (Dabrio et al., 2000). Comparing sediment-accumulation rate changes between estuaries, however, has limitations for quantifying impacts of increasing accommodation on depositional environments because flooding surfaces are time transgressive and depending on where sediment-accumulation rates are measured along depositional dip and strike, the timing and magnitude of the observed change will be influenced. Cronin et al. (2007) identified two marsh flooding events within the fill of Chesapeake Bay between 8900 and 8800 cal yr BP and 8200 and 7600 cal yr BP. The bay-head delta back-stepping events recognized in 5 estuaries along the northern Gulf of Mexico, the decrease in sedimentation rates observed in the East China Sea and the Gulf of Cadiz, and the flooding surfaces identified in the Chesapeake Bay, all occurring around 8200 cal yr BP, suggest a global event was the cause. It is likely that sea-level rise associated with the 8200 cal yr BP cold, dry, and windy period, documented in Greenland ice core records by Alley et al. (1997), triggered the observed estuarine changes.

Barber et al. (1999), Törnqvist et al. (2004), and Cronin et al. (2007) all suggest that the increase in the rate of sea-level rise around 8200 cal yr BP occurred as a result of a massive freshwater outflow event into the Labrador Sea. Although the outflow associated with this event was only 0.1 m in the northern Gulf of Mexico, the sea-level signal due to glacial isostatic adjustment over the same period was decelerating from about 1.0 to 0.5 m/century (Kendall et al., 2008). Rodriguez et al. (2005), Greene et al. (2007), and Simms et al. (2008) mapped low-gradient antecedent topography, interpreted as fluvial terraces, at −14 m in Galveston and Mobile bays, and −18 m in Corpus Christi Bay, respectively. Inundation of this low-gradient topography likely exacerbated the back-stepping event in these Gulf of Mexico estuaries.

The northern Gulf of Mexico estuaries did not respond instantaneously to the rapid sea-level rise. In addition to dating errors, each fluvial system and estuary had its own response time to a climatic or eustatic change, no matter how rapidly that change occurred (Blum and Törnqvist, 2001). For example, an estuary with a low sediment supply/accommodation creation ratio should have an immediate and significant response to an increased rate of sea-level rise, whereas an estuary with a high sediment supply /accommodation creation ratio should have either a subtle or no response (Cooper, 1993). Unfortunately, due to the error bars associated with radiocarbon dating, as well as the availability of material for dating, any differences in timing of the event among the five estuaries are indistinguishable.

6. Conclusions

Similar to other coastal areas, bay-head deltas across the northern Gulf of Mexico back stepped in response to the rapid increase in sea-level rise at 8200 cal yr BP. The amplitude of the sea-level rise anomaly was only 0.1 m in the northern Gulf of Mexico, although this was in addition to a background rate of sea-level rise that was decelerating from ~1.0 to 0.5 m/century. This rate is only about 13% of the modeled rate of rise experienced by the Gulf of Cadiz or the East China Sea (Kendall et al., 2008), where documented impacts to depositional environments are larger than what is presented here (Dabrio et al., 2000; Hori et al., 2007) and >150% of the modeled rate of rise experienced by Chesapeake Bay (Kendall et al., 2008), where documented impacts to marshes were likely smaller than what is presented here (Cronin et al., 2007). Given the episodic nature of ice-sheet retreat, there likely have been other episodes of punctuated rise that have gone unrecognized. In the Gulf of Mexico estuaries, there were flooding events similar to the 8200 cal yr BP event at ~9800–9500, ~8900–8500, ~7900–7500, and ~7400–6800 cal yr BP that appear to have been contemporaneous but neither corresponding sea-level rises nor glacial events have been recognized (Anderson et al., 2010). The estuarine impacts of the rapid sea-level rise at 8200 cal yr BP across the northern Gulf of Mexico may be an appropriate analog for impacts resulting from future sea-level rise (Overpeck and Weiss, 2009; Vermeer and Rahmstorf, 2009). Regardless of what future rates of sea-level rise will be, our results indicate that bay-head deltas, especially those characterized by low-gradient shorelines, are an especially sensitive estuarine component to low-amplitude sea-level rise events.

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